

David Jiang

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EDUCATION

University of Waterloo

Honours Computer Science (co-op)

200 University Ave. West, Waterloo, ON

Expected Graduation Date: Apr. 2029

- **GPA:** 3.98
- **Awards:** 95th percentile on the Canadian Computing Competition, Distinction on the Euclid Math Contest
- **Relevant Coursework:** Algorithm Design, Memory Management, Object Oriented Programming, Data Structures and Algorithms, Compilers, Linear Algebra, Calculus

SKILLS

Languages: JavaScript, TypeScript, Python, C, C++, Java, R, SQL, JSON, HTML, CSS, SCSS

Development Frameworks & APIs: React.js, React Native, SvelteKit, Google Firebase API, Node.js

Tools: Git, GitHub, VS Code, Google Cloud, Linux, Bash, Vite, Tailwind, SQLite3, NumPy, TensorFlow, PyTorch

PROJECTS

Full-stack Task Planning Web Application ([Link](#)) | *SvelteKit, Google Firebase API*

[Source code](#)

- Designed and implemented a **full-stack** task management system with an interactive UI, gaining **30+ active users** in under a year
- Engineered **Firebase Authentication** for secure login and **real-time database integration** for efficient state management
- Optimized front-end rendering and **state management using Svelte stores**, ensuring a highly responsive user experience

Python Instagram/Tiktok Clip Generator | *Python, Whisper, FFmpeg*

[Source code](#)

- Developed an **automated video processing pipeline** using Python, **cutting manual editing time by 95%+**
- Integrated **Whisper API** for high-accuracy subtitle generation, achieving **90%+ transcription precision**
- Automated video editing with MoviePy and FFmpeg, enabling **rapid production of over 50+ short-form clips** during testing

Read Aloud Chrome Extension | *Manifest V3, JavaScript, SpeechSynthesis API*

[Source code](#)

- Engineered a Chrome extension leveraging **JavaScript, Manifest V3, and SpeechSynthesis API** to convert on-screen text to speech, enhancing web accessibility
- Designed and deployed an efficient UI for user interaction, tested across **multiple web environments**

EXPERIENCE

Coding Club President

Oct. 2022 – Jun. 2024

St. Robert Catholic High School

Thornhill, ON

- Spearheaded and organized Python and **competitive programming** workshops, engaging **150+ students** and driving a **50%+ increase in Canadian Computing Competition** participation
- Designed and executed **algorithmic problem-solving** sessions, enhancing students' proficiency in **data structures** and logic
- Led development of internal coding events to simulate real-world technical interviews and improve problem-solving skills

C++ and Web Development Course Designer and Instructor

Jul. 2022 – Aug. 2023

Toronto Internationally Educated Individuals Association

Markham, ON

- Authored and delivered **20+ interactive coding lectures** covering C++ concepts and web development
- Built **web applications and C++ programs** as live demos to reinforce concepts and practical applications
- Built and maintained professional relationships with students, parents, and association director, resulting in being **personally invited back to teach in subsequent years**

FIRST Robotics Team Programmer

May 2023 – Feb. 2024

Ram's Robotics

Thornhill, ON

- Engineered core robotic movement algorithms using **OpenCV** and **RoadRunner**, enabling precise autonomous path planning
- Contributed to **software integration and debugging**, achieving **1st place** finishes in two qualifying competitions and **2nd place provincially**